

General Entomology

ZOO 1203

Lecture 12

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Lesson Objectives

After completion you should be able to;

- Explain the insect collection and preservation methods
- Discuss when we have to do an insect collection
- Practice collection and preservation techniques (practical)

Insect collection techniques

- Location
- Time of the day
- Number of people attending the collection
- Equipment

Collection equipment/ techniques

1. Insect nets
2. Beating tray/net
3. Aspirators
4. Berlese funnel
5. Winckler bag
6. Light trap
7. Malaise trap
8. Window trap
9. Pitfall trap
10. Baits and lure
11. Emergence trap



Insect nets



Beating Tray



Aspirator

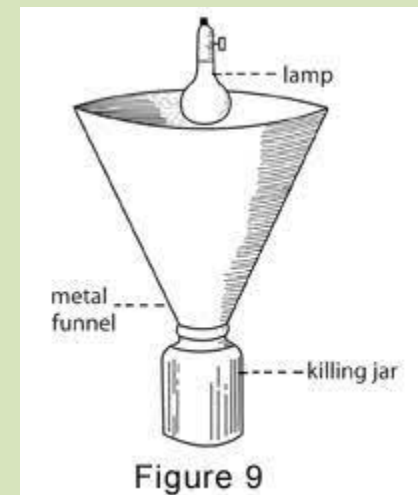




Berlese funnel



Winckler bag



Light trap



Malaise trap



Window trap



Pitfall trap



Emergence trap

Preserving insects

- Most adult insects are pinned or mounted and stored dry
- But adults of some orders and soft-bodied immature insects (eggs, larvae, nymphs, or pupae) are preserved in vials of 70 % - 80 % ethanol (ethyl alcohol)
- Or mounted on a microscopic slide
- Pupal cases, cocoons, waxy coverings and exuviae may be kept dry
- And either pinned, mounted on cards or points or, if delicate, stored in gelatin capsules or in preserving fluid.
- Two methods; dry preservation and wet preservation

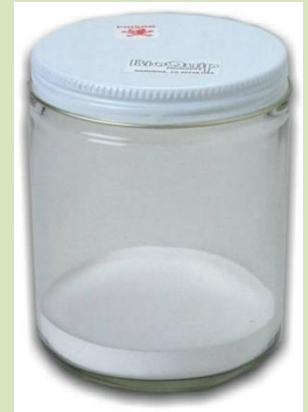
Dry preservation

Killing and handling prior to dry mounting

- best killed either using a killing bottle/jar/tube containing volatile poison
- Or in a freezer
- When freezing it is important to use an airtight container to prevent the drying out
- Freezing should be done at least 12 – 24 hours
- Few insects do not freeze easily and may not be killed easily in a normal household freezer

- Frozen insects have the risk of breaking. Thus must handle carefully
- The safest and most readily available liquid killing agent is ethyl acetate
- It is not dangerous unless directly inhaled
- But can use chloroform and cyanide but not recommended due to toxicity
- Killing containers are made by pouring thick mixture of plaster of Paris and water into the bottom of the jar/tube to a depth of 15 – 20 mm.

- The plaster must be completely dried before use.
- A small amount of ethyl acetate is poured onto the plaster and it will be absorbed by the plaster.
- Then it can be covered by a tissue
- Killing bottle should be kept cleaned
- Insects should be removed as soon as they die to avoid color loss
- Some insects like weevils may take many hours to die in ethyl acetate vapour



- Moths and butterflies should be killed separately
- Better to keep the dead insects under humid condition for 8 -24 hours (depending on the size of the species) to loosen their stiffed muscles/ until they have relaxed.
- Mounting must be done while they are fresh
- It is also important to remove the gut and other internal organs of large insects or gravid females (especially grasshoppers, cockroaches, mantids, stick insects and very large moths)

- Can do this by gutting, i.e. making a slit along the side of the abdomen (in membrane between the terga and sterna) using a fine, sharp scissors and removing the body content.
- Then a mixture of 3 parts of talcum powder and 1 part of boracic acid can be dusted into the body cavity
- In large insects body cavity may be stuffed carefully using cotton wool.

Wet preservation

- Most eggs, pupae, larvae, nymphs and soft bodied insects are preserved in liquid
- Because drying usually causes them to shrivel and rot
- Commonly used preservation liquid for the long-term storage of insects is ethanol (ethyl alcohol) mixed in various concentrations
- Usually the concentration lies between 75 – 80 %
- However aphids and scale insects are often preserved in lactic alcohol (two parts of ethanol and one part of lactic-alcohol)

- Because it will prevent them from becoming brittle
- Most immature insects will shrink and pigmented ones will discolor if placed directly onto ethanol
- Thus immature and soft bodied insects and specimens intended to study internal structures must be put onto fixatives solution prior to wet preservation



- Some commonly used fixatives are;
KAA – 2 parts glacial acetic acid, 10 parts 95% ethanol and
1 part kerosene

Carnoy's - 1 parts glacial acetic acid, 6 parts 95% ethanol
and 3 part chloroform

FAA - 1 parts glacial acetic acid, 25 parts 95% ethanol, 20
parts water and 5 part formalin

AGA - 1 parts glacial acetic acid, 6 parts 95% ethanol, 4
parts water and 1 part glycerol

Or simply by using 75% ethyl alcohol or isopropyl
alcohol

- Each specimen or collection should be stored in a separate glass vial or bottle that is sealed to prevent evaporation
- Data labels should be with the vials without letting it separated from the specimens
- Vials can be stored in racks or can be stored inside a larger jar containing ethanol to prevent evaporation.
- Even the best preserved specimens are a little or no scientific value if it is not labeled properly
- There must be a permanently printed label associated with the specimen

- The information must be recorded in the field note book at the time of collection
- Label should include;
 - location
 - Co-ordinates using GPS
 - Elevation – derived from a map rather than GPS
 - Date –sequence of day in Arabic numerals; month preferable in abbreviated letters (Feb – February);and year. Ex: 2 Apr 1999 or 2 Apr 99.
 - Collector's identity
 - Collection method

- In another label taxonomic identification can be included or it can be done in the same label too
- It is important never to discard the previous label as it might lead to lose useful evidence
- If the specimen is used in taxonomic description better put a label providing that information
- Labels must be print on high quality acid-free paper using permanent ink which can be provided now by high quality laser printer

Pinning, staging, pointing, carding, spreading and setting

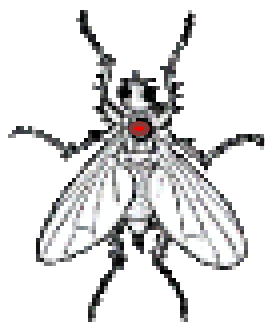
- Specimen should be mounted when their legs and wings are freely movable rather than stiff or dry
- Use entomological macropins
 - These are stainless steel pins,
 - mostly 32 mm long
 - random thickness
 - with a solid head
- Never use dressmakers pins for mounting insects



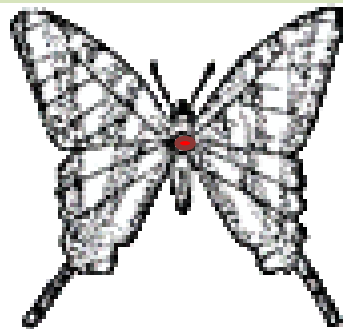
- There are three methods of mounting; direct pinning, micropinning/double mounting/staging, pointing
- Choice of the appropriate method depends on the kind of insect and its size as well as the purpose of mounting
- specimens $< \frac{1}{4}$ " mounted on a triangular points
- Specimens $> \frac{1}{4}$ " pinned with #2 or #3 pins
- Large robust specimens pinned with #7pins

Direct pinning

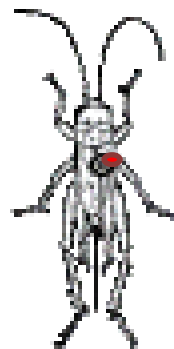
- Inserting a macropin of appropriate thickness for the insects' size directly through the insects' body
- The correct position of the body varies among insect orders
- Specimen should be positioned about three-quarters of the way up the pin with at least 7 mm protruding above the insect
- Specimen is held in the desired position on a polyethylene foam until it dries
- A dessicator or other artificial drying methods can be used in humid climate
- But oven temperature should be between 35 – 40 °C (this also depends on the specimen size)



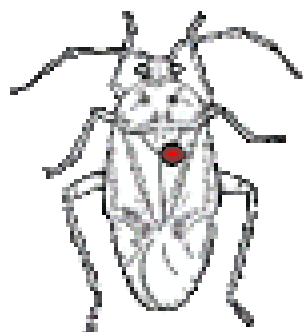
Flies &
Bees



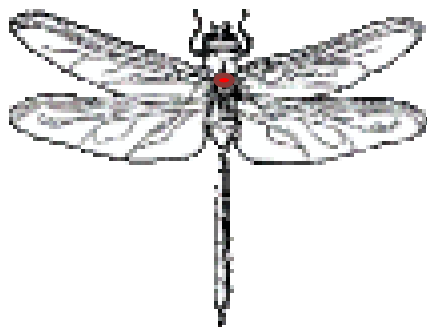
Butterflies &
Moths



Grasshoppers &
Crickets



True Bugs

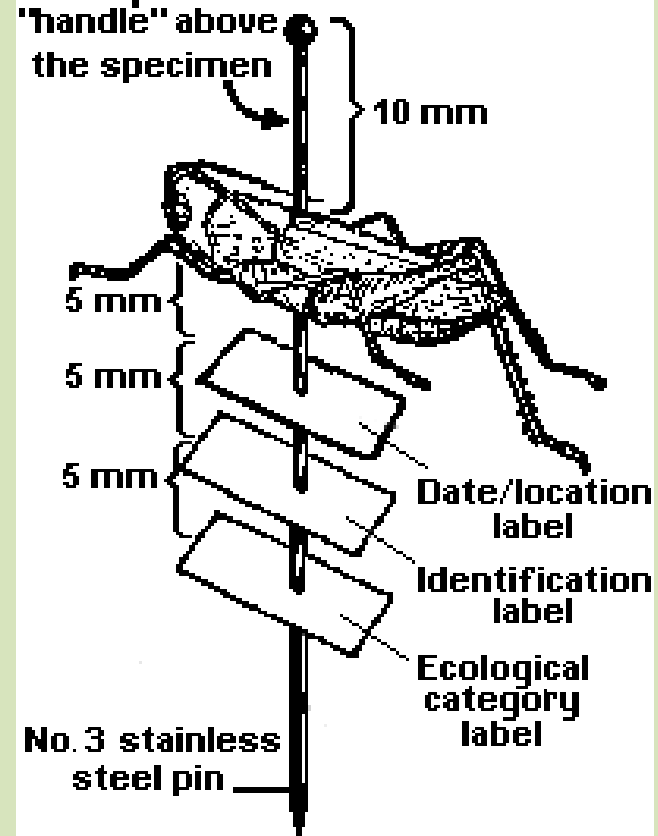


Dragonflies &
Damselflies



Beetles &
Weevils

Leave a 10 mm
"handle" above
the specimen

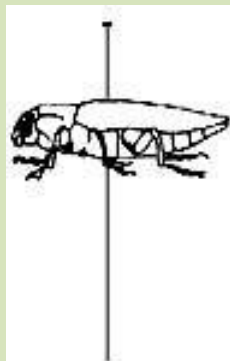


No. 3 stainless
steel pin

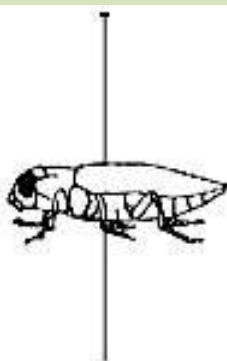
Date/location
label

Identification
label

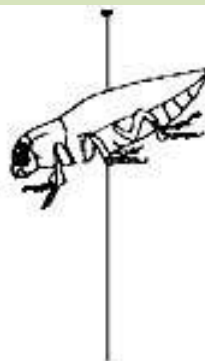
Ecological
category
label



Right



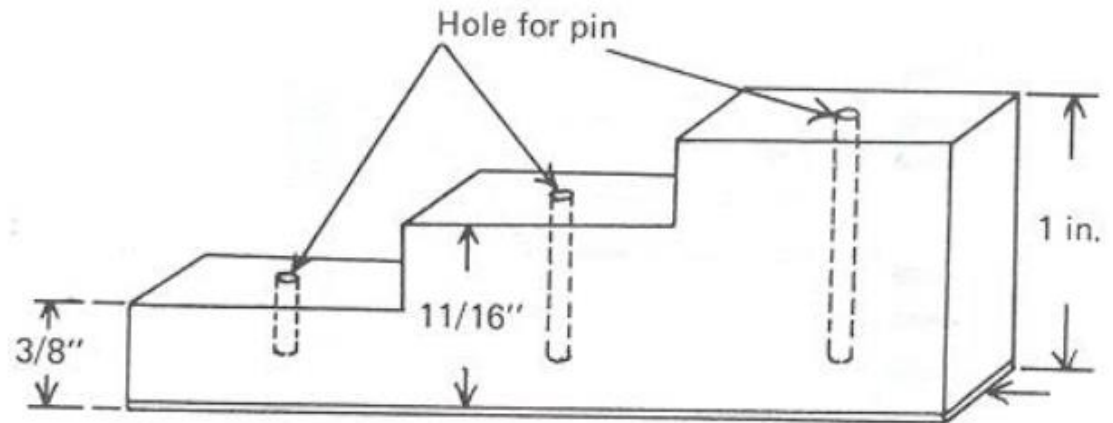
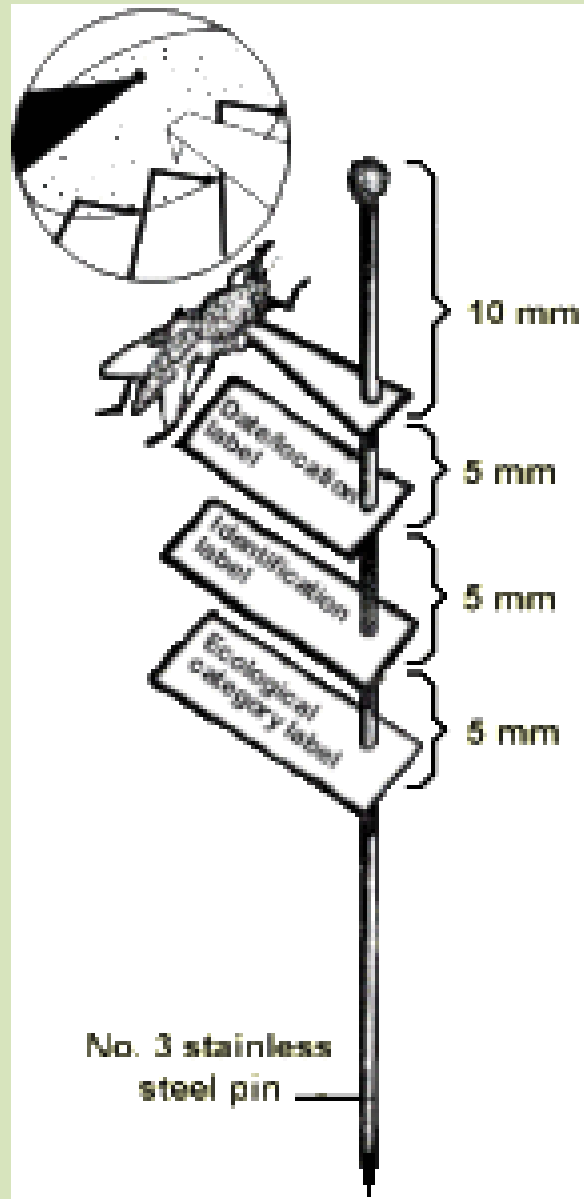
Wrong



Wrong

How to pin an insect →





Pinning step block used to regulate the height of specimens and labels on an insect pin

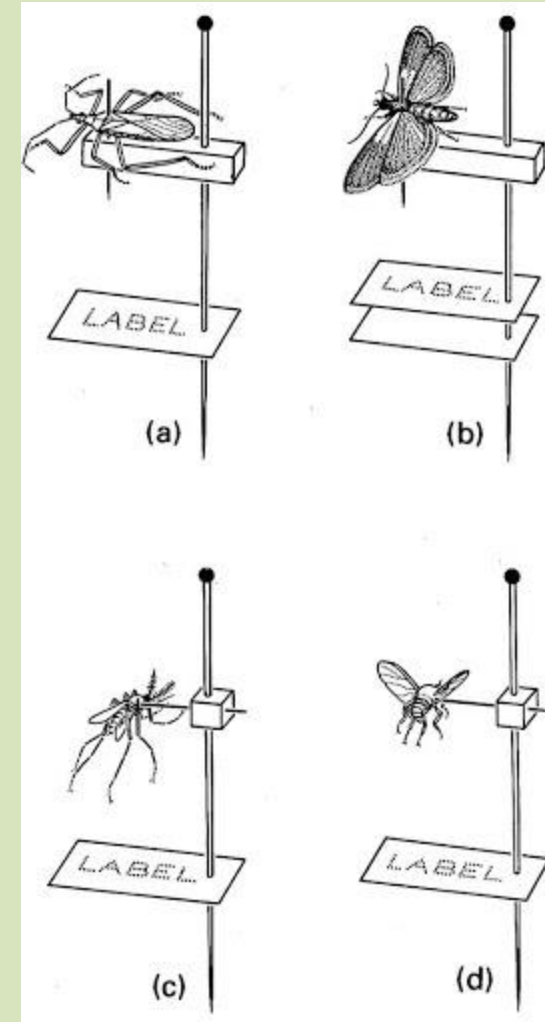
Mounting small insect →



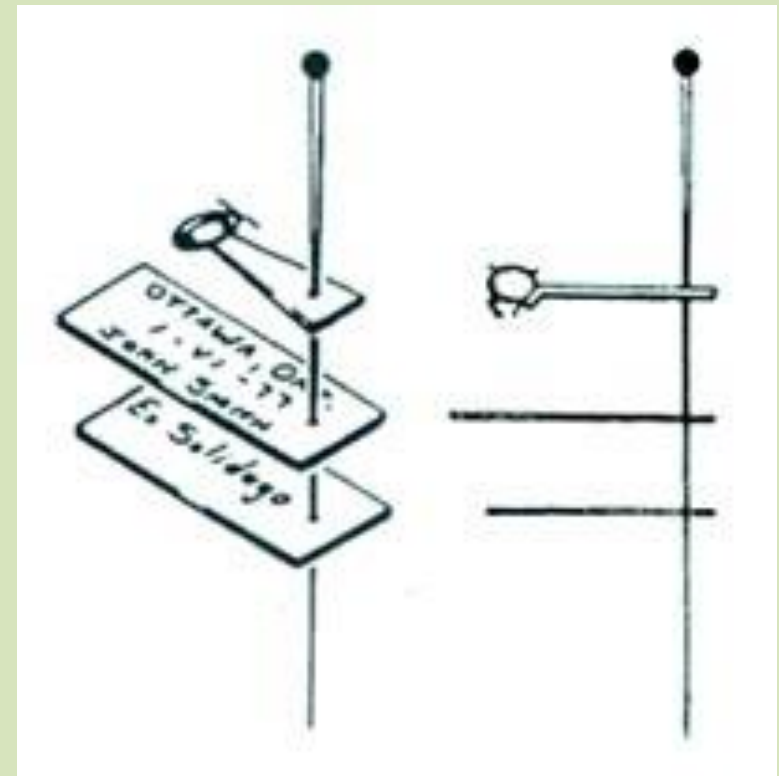
Micropinning/ staging/ double mounting

- Used for many small insects
- Involves pinning the insect with a micropin to a stage that is mounted on a macropin
- Micropins are very fine, headless, stainless steel pins from 10 – 15 mm long
- Stages are small square or rectangular strips of white polyporus piths or synthetic equivalents.
- Micropins are inserted in the same position as used in macropinning

- Small moths and wasps are mounted with the head facing away from the macropins and their bodies parallel to the stage
- Whereas beetles, bugs and flies are pinned with their bodies at right angles to the stage
- Some very small and delicate insects that are difficult to pin, such as mosquitoes and other small flies, are pinned to cube mounts.



- Pointing is used for small insects that would be damaged by pinning and very small sclerotized, small to medium size insects; especially weevils and ants whose cuticle is too hard to pierce with a micropin and
- mounting small specimens that are already dried
- points are made from small triangular pieces of white cardboard each point is mounted on a stout macropin
- it is inserted centrally near the base of the triangle
- insect is then glued to the tip of the point using a minute quantity of glue the head of the insect should be to the right
- tip should be contacted on the ventricle side of the thorax below the wings

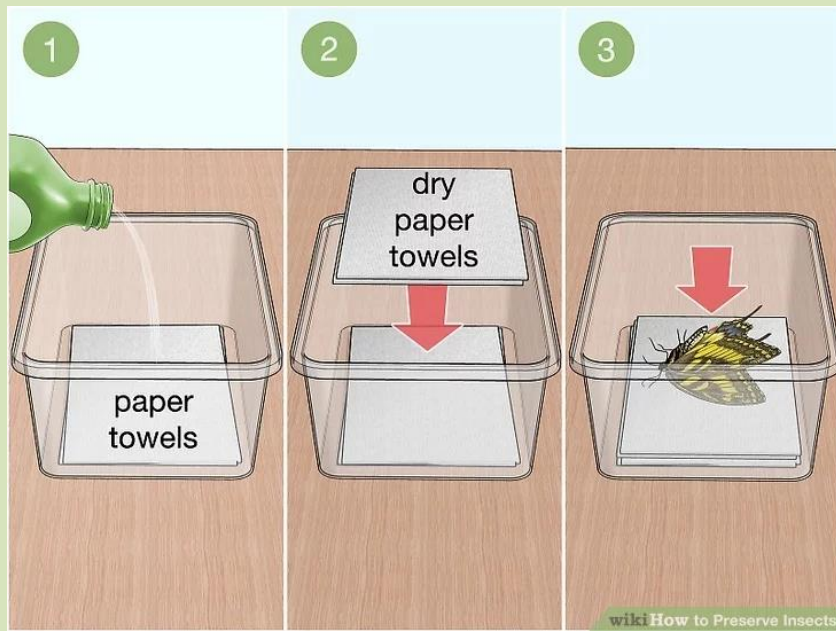


Spreading and setting

- Wings and antennae are important features used when identifying insects
- Spreading involves holding the appendages away from the body while the specimens are drying
- Legs and antennae can be held in a semi-natural positions with pins
- Wings can be opened and held out horizontally on a setting board using pieces of tracing paper, etc.

- Setting boards can be constructed using polyethylene foams
- Insects must be left to dry thoroughly before removing the pins







THANK YOU!

